Duy (Simon) Pham

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U.S. Citizen | Eligible for DoD Security Clearance

EDUCATION

University of Massachusetts Boston, College of Science & Mathematics Graduation: December 2027

B.S. Electrical Engineering, Minor in Computer Science

- GPA: 3.60 / 4.00 | Dean's List: 2023–2026
- Relevant Coursework: Signals & Systems, Fields and Waves, Electronics I & II, Antenna Design, Digital Systems, Probability & Random Processes

TECHNICAL SKILLS

RF & Analog Design: Software Defined Radios, AM/FM receivers, Op-amps, Filters, Microwave Circuits, Sensors, BJTs, MOSFETs, PCB Design

Programming & Software: Python, C/C++, MATLAB, LTspice, Linux, Bash, LaTeX, KiCAD, CAD Feko

Lab & Measurement: Oscilloscope, Function Generator, Multimeter, Spectrum Analyzer, Vector Network Analyzer

Embedded Systems: Raspberry Pi, STM32, Arduino, Nvidia Jetson, ESP32

WORK EXPERIENCE

Electrical Engineering Intern May 2026 – Present

RMF Engineering Inc.

- Assisted in the design of building electrical power systems, including distribution panels, lighting circuits, and branch circuits for project documentation
- Produced AutoCAD/Revit electrical drawings, including panel schedules and power distribution diagrams for construction drawing packages
- Performed electrical load calculations and voltage drop analysis for commercial building electrical systems during early design phases
- Collaborated with multidisciplinary engineering teams to coordinate electrical infrastructure with HVAC and architectural layouts

Undergraduate Research Assistant Jan 2026 – Present

Ubiquitous Communication and Networking (UCaN) Lab – Distributed Spectrum Sensing Algorithm Models

- Deployed and Implemented a multi-node wireless network via Raspberry Pis, collecting RSSI and throughput measurements for ML Algorithm Training
- Automated data collection and controlled experimental parameters XMLRPC, ZMQ Python scripts, implementing on Raspberry Pi node networks
- Modified Distributed Spectrum Sensing Algorithm submodels, increasing consensus accuracy to ∼95% and improving decision speed by ∼20%
- Integrated Linux-based Testbed Controller (Ubuntu) to support Software Defined Radio spectrum sensing experiments and Raspberry Pi control
- Developed MATLAB probability density function models to characterize detection thresholds, reducing false detection rates in the DSS algorithm
- Wrote and Deployed Bash scripts for testbed controller to Raspberry Pi, to transfer Python scripts, CSV files, and to save collected experimental data

Firmware & Embedded Systems Engineer Feb 2026 – May 2026

UMass Boston Robotics & Engineering Club – STM32 Drone Health Monitoring System

- Built and deployed in C a functional project prototype using Arduino, successfully integrating sample ML algorithm into PCB flight controller
- Ran 30+ motor health tests for Machine Learning Algorithm Training, using real time collected voltage, current, temperature, and RPM measurements
- Reduced false drone health alerts by 25%, maintaining 97% Drone Health Accuracy through algorithm training and tuning, validated with 300+ tests
- Deployed C code for STM32 microcontroller, to process ESC telemetry and monitor drone health in a local Python based UI with 10 ms updates

Undergraduate Teaching Assistant June 2026 – Aug 2026

UMass Boston Signals & Systems and Probability & Random Processes Courses

- Supported students through office hours, homework review, exam preparation, and reinforcement across engineering coursework
- Assisted students with Fourier transforms, convolution, sampling theorem, and probability distributions through technical problem-solving support

PROJECTS

AM Radio Receiver June 2025 – Nov 2025

- Implemented bandpass filter receiving 680 kHz, 850 kHz, and 1510 kHz radio waves, verifying real time audio from Boston radio stations
- Optimized RC filters lowering noise floor by ∼15% and reducing noise by ∼5 dB for clearer recovered speech, validated over 30+ test trials
- Designed and deployed a diode based envelope detector to rectify and demodulate AM audio signal from varying radio station carrier waves
- Designed, built, and tuned an LM386 audio amplifier stage with variable gain (5x–20x) for user controlled volume adjustment

Earth-Moon-Earth Antenna Feb 2026 – March 2026

- Modeled and analyzed a radar link budget utilizing the radar range equation to successfully guarantee a minimum Signal-to-Noise Ratio of 20 dB
- Calculated extreme free-space path loss (P_{loss} = -270.9 dB) at a 1.3 GHz carrier frequency over a 770,000 km dual-hop propagation path
- Derived system noise power (P_n = -144 dBm) using the kTB thermal noise formula across an assumed 1 kHz receiver bandwidth
- Determined optimal dimensions by evaluating aperture efficiency ($\eta_a = 0.6$) and an antenna gain of 43.45 dBi to specify a 14.1 meter dish diameter

CONFERENCES

NEWSDR Boston 2026 — New England Workshop on Software Defined Radio 2026

MassRobotics Boston 2026 — Form & Function Robotics Challenge Competition 2026